altairsim — Quick Start

2026-10-24 · 51399ea

Boot CP/M with one command. From the folder you unzipped this into:

```
$ ./altairsim examples/cpm/cpm22-buffered.toml
```

On **Windows**, the path is spelled with backslashes and the program is `altairsim.exe`:

```
> altairsim.exe examples\cpm\cpm22-buffered.toml
```

On **macOS**, start the program from a terminal, not with a double-click. Finder opens a plain executable as a document, and macOS refuses that even when the package is signed and notarized. If you got the package with a browser and macOS still blocks the first start, remove its quarantine mark. Type this once for each download:

```
$ xattr -dr com.apple.quarantine ./altairsim
```

If macOS shows a message about the program, press **Done**, not **Move to Trash**.

That is the whole of it. The machine file boots itself — you do not type a `BOOT` command, because an Altair had none — and a few seconds later you are in CP/M:

```
56K CP/M 2.2b v2.3
For Altair 8" Floppy

A>
```

It is 1977. Type `DIR` to see what is on the disk:

```
A>DIR
```

An assembler, a debugger, Microsoft BASIC, a text editor, and a game or two are on there. Run one — `A>MBASIC` starts Microsoft BASIC.

The three keys to remember

You are talking to CP/M, the software the CPU is running. Two prompts share the window: `A>` is CP/M, and `altairsim>` is the simulator's own monitor. You move between them with these:

`^E` (Control-E)	Stop the CPU and return to the `altairsim>` monitor. The machine stops exactly where it stood and loses nothing.
`RUN`	Start the CPU running again. It picks up where it left off.
`QUIT`	Done. (`Q` will do. There is no `EXIT`.)

`^C` is *not* how you get out — that belongs to CP/M, which warm-boots on it. Use `^E`.

The disk is real — there is no undo

The disk is mounted read/write, because that is what a machine with a disk in it is. Anything you do in CP/M happens to the file on your host, and nothing keeps a spare copy. Two ways to stay safe:

- **Look around read-only.** Add `RO` when you mount, and every write is refused at the controller so the file cannot change.
- **Copy the folder first** when you mean to write. It is self-contained and boots from anywhere:

```
$ cp -R examples/cpm my-cpm
$ ./altairsim my-cpm/cpm22-buffered.toml
```

One trap worth knowing: this CP/M **buffers a whole track** and only flushes it the next time it reads the console, so **get back to the `A>` prompt before you quit or copy the image**, or the last write never lands.

Where the files come from

You named **one** path — the machine file — and the simulator found the disk on its own. One rule covers all of it:

- **The machine file you name on the command line** is found from the folder you are in — your shell. That is the one path you hand to the shell, before any machine exists.
- **Everything after resolves against the machine's own folder** — the directory that `.toml` lives in. The disk it mounts, and the `MOUNT`, `LOAD`, `SAVE` and `DO` you type at the `altairsim>` prompt, all come from there. `cpm22-buffered.toml` mounts `cpm22b23-56k.dsk` from *beside itself* — and if you type `MOUNT dsk0:drive1 cpm22b23-56k.dsk` you get that same file, from that same folder.

For example, from the folder you unzipped into:

```
$ pwd
/home/you/altairsim
$ ./altairsim examples/cpm/cpm22-buffered.toml
```

`examples/cpm/cpm22-buffered.toml` is found from where you are; the `cpm22b23-56k.dsk` it mounts — and anything you later type at the prompt — is found in `examples/cpm/`, beside the machine.

That is why every folder under `examples/` boots wherever you copy it: the disk travels with the `.toml` that names it, and so does everything you do to that machine. `altairsim` never changes your working directory.

To see it, ask the machine: `SHOW PATHS` at the `altairsim>` prompt prints the base directory — and the hostbridge sandbox, the one folder kept separate:

- **the base directory** — the machine's own folder; what a machine file mounts and what you type both resolve against it;
- **the sandbox root** — the one folder the hostbridge file-transfer board may reach, and cannot escape.

The leading `./` means *the `altairsim` in this folder* — a fresh unzip is not on your `PATH`, so you point at it. Copy the program into a directory on your `PATH` (`/usr/local/bin` on macOS or Linux) and you can drop the `./` and type just `altairsim` from anywhere, including from inside an example folder: `cd examples/cpm && altairsim cpm22-buffered.toml`.

No disk? Boot a tape instead

Every folder under `examples/` is a complete machine with its media already inside it, so any of them comes up the moment you name it:

```
$ ls examples/
$ ./altairsim examples/basic4k/basic4k.toml
```

`examples/basic4k/` toggles in the MITS bootstrap by hand and loads **Altair 4K BASIC** off a cassette — the machine that shows you what an Altair actually was. Each folder carries its own `README.pdf` describing what it is and what to type. More examples, and their documentation, are at https://altairsim.com.

Where to go next

`altairsim-manual.pdf` is the full User Manual — the same quick start above at more length, then the machines, the disks and tapes, the S-100 boards, and the file-transfer utilities in `hostbridge/`. `altairsim-monitor.pdf` and `altairsim-debugger.pdf` cover the `altairsim>` prompt and the built-in debugger. If you are coming from AltairZ80 (SIMH) or z80pack, read `migrating.pdf` first.